Somnath Chattaraj

+91 9382054930 | somnathchattaraj5@gmail.com | LinkedIn | GitHub | Portfolio | Codeforces | LeetCode

Education

Jadavpur University — Kolkata, West Bengal
*Bachelor of Engineering in Information Technology; CGPA: **8.28/10*** — *Aug. 2023 – Jun. 2027*

Experience

Google — Bengaluru, Karnataka
Software Engineering Intern — *May 2026 – Aug. 2026*

- Contributed to **Android's Recents View** within the Motion Hero Moment team, working across the **Android framework** and user-interface rendering pipeline.
- Implemented **3 core interactions**—zoom for the highlighted task and dynamic task-card scaling—and integrated transition and snap animations for smoother task switching.
- Optimized rendering and animation behavior in the Recents pipeline, **reducing dropped frames by 4%** and improving responsiveness during task switching.
- Developed and integrated **1,000+ lines of production code** for Android framework features in collaboration with **2 cross-functional teams**.
- Created and executed **20+ test scenarios**, resolved defects, and incorporated feedback from **100+ code reviews** to deliver reliable production-quality changes.

Projects

Incident Detection & Auto-Remediation ↗ | *TypeScript, Express.js, BullMQ, Prometheus, ChromaDB, LLMs*

- Built an **SRE platform** for autonomous incident detection, root-cause analysis, and automated code remediation using **LLM- and RAG-based workflows**.
- Developed **event-driven backend pipelines** with **Express.js, BullMQ, and Redis**, processing up to **20 anomaly events per minute** in local load tests.
- Integrated **Prometheus** monitoring across **12 service metrics** and built a **ChromaDB-backed auto-patching engine** to retrieve code context, generate fixes, and automate Git-based remediation workflows.

Automation Workflow Platform ↗ | *Kafka, Express.js, Prisma, Webhooks*

- Engineered an **event-driven automation platform** for defining and executing custom workflows triggered by incoming **webhooks**.
- Used **Kafka** for asynchronous event processing and **Prisma** for persistent workflow and application data, supporting workflows with up to **5 configurable steps**.
- Designed a modular **microservices-based backend architecture** with **4 independently maintainable services** for workflow execution, event handling, and data persistence.

ECOX ↗ | *Solidity, Avalanche, Chainlink, MERN, LangChain, Gemini API*

- Developed a web application and secure **REST APIs** for creating, verifying, and trading **carbon-credit NFTs** on **Avalanche**.
- Implemented **8 Solidity smart-contract functions** for NFT issuance, ownership transfer, verification, and marketplace interactions, with **Chainlink automation** for on-chain task execution.
- Integrated **Gemini API**-based certificate analysis and validation, reducing manual verification to a single API-driven workflow.

Technical Skills

Languages: C, C++, Java, Kotlin, Python, TypeScript, SQL
Backend & Distributed Systems: Node.js, Express.js, REST APIs, Kafka, Redis, BullMQ, PostgreSQL
Cloud & Reliability: Docker, Kubernetes, Prometheus, Google Cloud Platform, Monitoring
Systems & Developer Tools: Android Framework, Git, Debugging, Performance Optimization

Achievements

- **Codeforces: Specialist**.
- **LeetCode:** Solved **700+ algorithmic problems**.
- Ranked **16th nationally** and **143rd globally** in IMC Trading Prosperity 4, an international trading competition.
- Placed in the **top 8 among 400+ teams** at the East India Blockchain Summit Hackathon, IIT Kharagpur.